

Balancing Traffic Across Data Centres Using

LVS



In the previous article in this series ('Data Centre Redundancy', page 56, April 2009) we got an overview of the architecture associated with data centre redundancy. An important aspect of this is to make sure that the traffic is evenly distributed across all the data centres in order to fully utilise the available capacity. In this concluding article, we will discuss the use of LVS (Linux Virtual Server) to load balance the traffic across data centres.

The LVS (Linux Virtual Server) project was launched in 1998 and is meant to eliminate Single Point of Failures (SPOF).

According to the linuxvirtualserver.org website: "LVS is a highly scalable and available server built on a cluster of real servers, with the load balancer running on Linux. The architecture of the server cluster is fully transparent to the end user, and the users interact as if it were a single high-performance virtual server. The real servers and the load balancers may be interconnected by either a high speed LAN or by a geographically dispersed WAN."

The load balancer is the single entry point into the cluster. The client connects to a single known IP address, and then inside the virtual server the load balancer redirects the incoming connections to the server(s) that actually does the work according to the scheduling algorithm chosen. The nodes of the cluster (real servers) can be transparently added/removed, providing a high level of scalability. The LVS detects node

failures on-the-fly and reconfigures the system accordingly, automatically, thus providing high availability. Theoretically, the load balancer can either run IPVS or KTCPVS techniques for load balancing, but owing to a very high stability of IPVS, it is used in almost all the implementations I have seen. See the sidebar titled "IPVS v/s KTCPVS" for a brief note on the differences between the two. IPVS provides Layer 4 load balancing and KTCPVS provides Layer 7 load balancing (see the sidebar).

There are three load balancing techniques used in IPVS:

- LVS/NAT – Virtual Server via NAT
- LVS/TUN – Virtual Server via Tunneling
- LVS/DR – Virtual Server via Direct Routing

A brief overview of these techniques can be found in the sidebar titled 'IPVS Load Balancing Techniques'.

Since our real servers are located in two different data centres, we will be focusing on LVS/TUN.

Installing and configuring IPVS

Please note that the set-up explained here should only be used as a guideline and for an understanding of how IPVS works. Networking scenarios are different for every case and may demand extra reading and experimentation before getting a working set-up. My advice is that before trying this out in the field, make sure enough experiments have been done in the laboratory. Also, it is advisable to read through the documents in the *References* section at the end of the article.

IPVS v/s KTCPVS

IPVS or IP Virtual Server is an implementation of Layer 4 load balancing inside the Linux kernel. Layer 4 load balancing works on OSI Layer 4 (Transport Layer) and distributes requests to the servers at the transport layer without looking at the content of the packets.

KTCPVS or Kernel TCP Virtual Server is an implementation of Layer 7 load balancing in the Linux kernel. Layer 7 load balancing is also known as application-level load balancing. The load balancer parses requests in the application layer and distributes requests to servers based on the content. The scalability of Layer 7 load balancing is not high because of the overhead of parsing the content.